EXPERIMENT 13

Case Study on Limitations of Low-Code Platforms for a Complex Enterprise Use Case

Name: Jenell S G | Register Number: 240701212

Title

Case Study on Limitations of Low-Code Platforms for a Complex Enterprise Use Case – Enterprise E-Waste Collection and Recycling Management System.

Aim

To study the limitations of low-code platforms when they are applied to complex enterprise-level applications, using the EcoCycle application built in Zoho Creator as the starting point, and to understand when pro-code extensions or a hybrid architecture become necessary.

Objectives

- To describe how an enterprise-scale e-waste collection and recycling system is structured, from users to analytics.
- To map the standard low-code capabilities (forms, reports, workflows, data model, integrations) to the operations of the system.
- To examine eight core limitations of low-code at enterprise scale and relate each to what was seen in EcoCycle.
- To present a three-tier hybrid architecture (low-code application, pro-code extensions, enterprise systems) as the resolution.
- To judge when the choice between low-code, pro-code and hybrid development should be made.

Software / Tools Required

- Zoho Creator (web-based low-code application development platform)
- EcoCycle application (E-Waste Entry, Collection Request and Recycling Tracking forms and their reports)
- Web browser with an active internet connection

Introduction

Low-code platforms let applications be built mainly through visual tools instead of large amounts of program code. Applications are assembled from forms, reports and dashboards, and behaviour is defined through workflows and a small amount of scripting. This reduces development time and lets a wider range of people take part in building business applications.

Low-code suits standard business applications. As the users, data volume, process complexity, integrations and security demands of an organisation grow, the built-in capabilities of a platform may no longer be enough on their own. This case study examines such limitations by scaling up EcoCycle from the small test application of the project to an enterprise-wide e-waste network. The enterprise system described below is a hypothetical scale-up; the observations about EcoCycle itself are taken from the project report.

Case Study: Enterprise E-Waste Collection and Recycling Management System

EcoCycle collects e-waste entries through the E-Waste Entry form, calculates a priority score and level in a Deluge workflow, accepts collection requests, tracks recycling, creates a lead in Zoho CRM for each request, shows the position on a dashboard and offers a read-only Zia chat agent. Suppose a city or a state must now run an e-waste network covering many institutions, collection teams and recyclers, with a very large number of items each year. The system would then serve the following users and functions.

Category	Items
Users	Institutions and offices that report e-waste, collection staff, recyclers, e-waste managers and administrators
Data and operations	E-waste entries, collection requests, recycling records, hazardous-material and condition details, collection schedules and item history
Supporting functions	Priority alerts, request and status notifications, SMS notifications, QR-code identification of items and real-time collection tracking
Enterprise integration and insight	CRM integration, map and route services, regulatory reporting, analytics and reporting

Figure 13.1 shows the architecture of this system on a low-code platform, arranged as six groups connected from top to bottom.

USERS	Institution users • Collection staff • Recyclers • E-waste manager • Administrators
APP INTERFACE	E-Waste Entry form • Collection Request form • Recycling Tracking form • Reports • AI Agents page
LOGIC & WORKFLOW	Priority calculation • status defaults • past-date check • high-priority alert • collection scheduling • CRM lead creation
DATA MANAGEMENT	E-waste records • request records • recycling records • recycler and location master data • item history
EXTERNAL SYSTEMS	Zoho CRM • Google Maps • SMS gateway • QR-code service • regulator reporting • Creator REST API
ANALYTICS & REPORTING	Priority distribution • collection and recycling status • recycled share • management dashboard

Figure 13.1: Six-layer architecture of the enterprise e-waste management system

- **Users:** the people who report, collect, recycle and manage e-waste.
- **App Interface:** the forms, reports and chat page through which data is entered and viewed.
- **Logic & Workflow:** the Deluge rules that move an item from entry to collection and recycling.
- **Data Management:** the records in which all information is stored.
- **External Systems:** the services with which the application must exchange information.
- **Analytics & Reporting:** the charts and reports used by decision makers.

How a Low-Code Platform Can Support the System

Capability	Use in the e-waste system
Forms	E-Waste Entry, Collection Request and Recycling Tracking capture items, requests and recycling jobs.

Capability	Use in the e-waste system
Reports	All E-Waste, High Priority E-Waste, Collection Requests and Recycling Tracking Report list the records.
Workflows	Priority calculation, status defaults, date validation and the high-priority alert.
Data management	Stores records and performs standard create, read, update and delete operations.
User interfaces	Web screens with desktop, tablet and mobile views and a dashboard of charts.
Basic integrations	A Zoho CRM connection and a Custom API secured with OAuth2.
AI assistance	The EcoCycle Data Assistant, a read-only Zia chat agent.

The Eight Core Limitations at Enterprise Scale

As the system grows into a complex enterprise platform, a basic low-code application meets growing enterprise requirements and then enterprise-scale challenges, which lead to the eight limitations below.

1	2	3	4	5	6	7	8
Perfor-mance	Complex logic	Integra-tion	UX custom-isation	Security	AI / advanced tech	Vendor lock-in	Data transf or-mation

Figure 13.2: Limitations of low-code platforms at enterprise scale

1. Performance and Scalability

When items, requests and concurrent users grow very large, for example thousands of institutions reporting at the end of a year or after a clean-up drive, complex reports and charts over large data may respond slowly. The infrastructure is managed by the platform, so the options for tuning performance are limited. In EcoCycle no performance measurement was taken, and the dashboard reads directly from the forms.

2. Complex Business Logic

Rules such as weights that change by item type, regional hazardous-waste regulations, route and vehicle optimisation for collection teams, or multi-level approval of collection schedules are difficult to express in visual workflows and standard scripting. EcoCycle's priority score is a short if-else rule, and its factor weights are not yet documented; a richer rule set would need custom development.

3. System Integration

Connecting to mapping, SMS, QR-code, regulator and legacy asset-register systems can be challenging. Built-in connectors may not cover every system or data format, and real-time integrations may need custom interfaces. EcoCycle's Zoho CRM link works through a connection, but the Creator REST API integration, Google Maps and SMS notifications are still future work.

4. Extreme UX Customisation

Standard components provide common screens and layouts. A guided multi-step reporting wizard, a branded public portal or a mobile photo-capture experience for scanning item labels may exceed what standard components provide. EcoCycle uses plain single-column forms.

5. Security and Governance

Organisations need detailed access control, audit trails and retention rules for personal data. EcoCycle defines no roles, so every user who can open it sees all records, including requesters' e-mail addresses and phone numbers, and it stores personal contact details without a stated retention rule. Role-based

controls (Experiment 8) help, but fine-grained or regulatory requirements may need more than the platform provides.

6. AI / ML and Advanced Technology

Specialised models, such as estimating recoverable material from item photographs or predicting collection volumes, may need external services or custom code. EcoCycle's chat agent only reads data and its answers must be verified; the page itself advises users to review AI-generated responses.

7. Vendor / Platform Dependency

The application depends on the capabilities, pricing and supported integrations of the platform and on the plan or edition of the account. EcoCycle runs on a trial account that showed 13 days remaining; its connections and AI features are configured outside the forms and must be recreated if the application is moved.

8. Heavy Data Transformations

Large imports of asset registers, de-duplication of items across institutions and complex transformations for regulatory reports may be hard to handle within the data model and tools of the platform. EcoCycle's forms are related only by repeated values and not by lookups, and Collection Location is free text, so values such as A Block and A block already differ.

Relevance of each limitation to the case study

Limitation	Relevance to the e-waste system	Seen in EcoCycle
Performance & scalability	Growing items, requests and reports over history	Not measured (no figures reported)
Complex business logic	Region-specific scoring, scheduling and routing rules	Simple if-else score; weights not documented
System integration	Maps, SMS, QR codes, regulators, asset registers	CRM built; REST API planned
UX customisation	Portals and mobile capture for collectors	Plain single-column forms
Security & governance	Protecting personal contact data; audit trail	No roles; all records visible
AI / advanced technology	Forecasting and image-based assessment	Read-only chat agent; answers to be verified
Vendor dependency	Reliance on the platform for all features	Trial account; connections tied to the account
Heavy data transformations	Bulk imports, de-duplication, regulatory reports	No lookups; inconsistent location text

Low-Code vs Pro-Code Requirements

Neither approach is universally better; the comparison below indicates the typical role of each.

Area	Low-code approach	When pro-code may be required
User interface	Standard screens, forms and layouts	Highly specialised or branded interfaces
Business logic	Visual workflows and simple Deluge scripts	Complex rules and advanced processing
Data management	Built-in data model and standard CRUD operations	Very large datasets and complex transformations

Area	Low-code approach	When pro-code may be required
Integrations	Connections and standard interfaces	Legacy, specialised or real-time systems
AI / ML	Platform features such as the Zia chat agent	Custom models and ML pipelines
Security	Role- and permission-based access control	Fine-grained controls and compliance beyond the platform
Scalability	Suitable where the platform handles expected load	High data volumes and demanding performance

Resolution Topology: Three-Tier Hybrid Architecture

Organisations can combine low-code development with pro-code extensions. Low-code builds the standard application quickly, while custom code, APIs and specialised processing handle what exceeds the platform. Figure 13.3 shows the three layers; information flows in both directions between them.

USERS	Institution users • Collection staff • Recyclers • E-waste manager • Administrators
TIER 1: LOW-CODE APPLICATION	EcoCycle forms, reports, dashboard, form workflows, CRM connection and chat agent (built on Zoho Creator)
TIER 2: PRO-CODE EXTENSIONS	Custom functions and REST API services for scoring rules, route optimisation, bulk imports and AI services (for example Zoho Catalyst)
TIER 3: ENTERPRISE SYSTEMS	Zoho CRM • map and route services • SMS gateway • regulator reporting • asset registers

Figure 13.3: Three-tier hybrid architecture

Why hybrid?

- The standard layer (forms, reports, dashboard, simple rules) keeps the speed and low cost of low-code.
- Pro-code extensions are added only for the requirements that exceed the platform.
- Business users can continue to maintain the forms and reports while developers maintain the extensions.
- The architecture can grow in steps, for example by adding the planned REST API integration first.

Decision guide

Situation	Choice
Standard forms, reports and rules for a small team	Low-code
Mostly standard needs with a few demanding parts, such as maps or bulk imports	Hybrid (low-code with pro-code extensions)
Highly specialised, high-volume or tightly regulated system	Pro-code or an enterprise platform

Result

The case study showed the major limitations that arise when low-code is applied to complex enterprise requirements: performance and scalability, complex business logic, system integration, extreme UX customisation, security and governance, AI / ML and advanced technology, vendor or platform dependency, and heavy data transformations. Several of them were already visible in the small EcoCycle application. It also showed how a three-tier hybrid architecture of a low-code application,

pro-code extensions and enterprise systems can address requirements beyond standard low-code capabilities.

Conclusion

Low-code platforms offer speed and simplicity for standard business applications, but their limitations become more significant as enterprise complexity increases. The trade-off is clear: low-code gives fast delivery, lower cost and wider participation in development, in exchange for ceilings in scale, logic, integration, customisation, governance and platform independence. For an enterprise e-waste system, the standard functionality can be built quickly with low-code, while demanding requirements in logic, integration, security, scale and advanced technology need pro-code extensions connected to enterprise systems. A hybrid architecture keeps the speed of low-code for the standard layer and adds the flexibility of custom code where it is needed. It is therefore important to understand the limitations of a platform before selecting a development approach.